

Yen-Chung Chen

PHD CANDIDATE

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Education

New York University

New York, NY, U.S.A.

DOCTOR OF PHILOSOPHY: BIOLOGY, GPA: 4.0/4.0

Sep. 2018 – Present

- Henry M. MacKracken Fellow
- NYSTEM Predoctoral Fellow

National Taiwan University

Taipei, Taiwan

DOCTOR OF MEDICINE, GPA: 3.45/4.0, GRE: V160/Q170/AW4.5

Sep. 2007 – Jun. 2014

Research Experience

Dr. Claude Desplan Lab, New York University

NY, USA

PHD CANDIDATE

June 2019 – Current

- Molecular logic of spatial patterning in the fly visual system

Dr. Jun-An Chen Lab, Institute of Molecular Biology, Academia Sinica

Taipei, Taiwan

RESEARCH ASSISTANT

August 2015 – August 2018

- Studying the roles of non-coding RNAs in diseases and development with patient derived iPS cells

Dr. Min-Chuan Huang Lab, National Taiwan University College of Medicine

Taipei, Taiwan

UNDERGRADUATE RESEARCH

January 2009 – August 2010

- Screening dysregulated glycosyltransferase expression in hepatic cancer

Publications

Isabel Holguera*, **Yen-Chung Chen***, Yu-Chieh David Chen, Felix Simon, Adelia Gaffney, Joey Rodas, Sergio Córdoba, Claude Desplan. “Temporal and Notch Identity Determine Neuropil Targeting Depth and Synapse Location in the Fly Visual System” Under review in *Nature Neuroscience*. 2025.01.06.631439 (2026)

Ben Jiwon Choi, **Yen-Chung Chen**, Claude Desplan. “Retinal Calcium Waves Coordinate Uniform Tissue Patterning of the *Drosophila* Eye” *Science*. (2025)

Rose Coyne, McKenzie Treese, **Yen-Chung Chen**, Cathleen Lake, Ojong Besong Tabi, Raghuvanshi Rajesh, Huzaifa Hassan, Hua Li, Claude Desplan, Mehmet Neset Özel. “Regulatory Logic of Neuronal Identity Specification in *Drosophila*” *bioRxiv: The Preprint Server for Biology*. 2025.09.01.673531 (2025)

Olivia V. Goldman, Alexandra E. DeFoe, Yanyan Qi, Nadav Shai. “A single-nucleus transcriptomic atlas of the adult *Aedes aegypti* mosquito” *bioRxiv*. (2025)

Félix Simon, Isabel Holguera, **Yen-Chung Chen**, Jennifer Malin, Priscilla Valentino, Claire Njoo-Deplante, Rana El-Danaf, Katarina Kapuralin, Ted Erclik, Nikos Konstantinides, Mehmet Neset Özel, Claude Desplan. “Establishment of terminal selector combinations in optic lobe neurons” *Nature Neuroscience (in press)*. (2025)

Justin T. Walsh, Ian P. Junker, Yu-Chieh David Chen, **Yen-Chung Chen**, Helena Gifford, Dawn Chen, Yun Ding. “High-Resolution Comparative Single-Cell Transcriptomics of Doublesex-Expressing Neurons Reveals Evolutionary Conservation and Diversity of Sexual Circuits in *Drosophila*” *PNAS (In revision)*. (2025)

Jennifer A. Malin, **Yen-Chung Chen**, Félix Simon, Evelyn Keefer, Claude Desplan. “Spatial Patterning Controls Neuron Numbers in the *Drosophila* Visual System” *Developmental Cell*. S1534580724001485 (2024)

Yu-Chieh David Chen, **Yen-Chung Chen**, Raghuvanshi Rajesh, Nathalie Shoji, Maisha Jacy, Haluk Lacin, Ted Erclik, Claude Desplan. “Using Single-Cell RNA Sequencing to Generate Predictive Cell-Type-Specific Split-GAL4 Reagents throughout Development” *Proceedings of the National Academy of Sciences* **120**(32), e2307451120 (2023)

Tai-Heng Chen, Shih-Hsin Chang, Yu-Fu Wu, Ya-Ping Yen, Fang-Yu Hsu, **Yen-Chung Chen**, Yang Ming, Ho-Chiang Hsu, Yi-Ching Su, Sheng-Tang Wong, Jui-Hung Hung, Shih-Hwa Chiou, Yuh-Jyh Jong, Jun-An Chen. “MiR34 Contributes to Spinal Muscular Atrophy and AAV9-mediated Delivery of MiR34a Ameliorates the Motor Deficits in SMA Mice” *Molecular Therapy - Nucleic Acids*. S2162253123000641 (2023)

Ee Shan Liao, Suoqin Jin, **Yen-Chung Chen**, Wei-Szu Liu, Maëliss Calon, Stéphane Nedelec, Qing Nie, Jun-An Chen. “Single-Cell Transcriptomic Analysis Reveals Diversity within Mammalian Spinal Motor Neurons” *Nature Communications* **14**(1), 46 (2023)

Tzu-Chiao Lu, Maria Brbić, Ye-Jin Park, ..., **Yen-Chung Chen**, ..., Hongjie Li. “Aging Fly Cell Atlas Identifies Exhaustive Aging Features at Cellular Resolution” *Science* **380**(6650), eadg0934 (2023)

Yen-Chung Chen, Nikolaos Konstantinides. “Integration of Spatial and Temporal Patterning in the Invertebrate and Vertebrate Nervous System” *Frontiers in Neuroscience* **16**, 854422 (2022)

Brianne A. Kent, Constance Holman, Emmanuella Amoako, ..., **Yen-Chung Chen**, ..., Tracey L. Weissgerber. "Recommendations for Empowering Early Career Researchers to Improve Research Culture and Practice" *PLOS Biology* **20**(7), e3001680 (2022)

Nikolaos Konstantinides, Isabel Holguera, Anthony M. Rossi, Aristides Escobar, Liébaut Dudragne, **Yen-Chung Chen**, Thanh N. Tran, Azalia M. Martínez Jaimes, Mehmet Neset Özel, Félix Simon, Zhiping Shao, Nadejda M. Tsankova, John F. Fullard, Uwe Walldorf, Panos Roussos, Claude Desplan. "A Complete Temporal Transcription Factor Series in the Fly Visual System" *Nature* **604**(7905), 316–322 (2022)

Seungjae Lee, **Yen-Chung Chen**, FCA Consortium, Austin E. Gillen, J. Matthew Taliaferro, Bart Deplancke, Hongjie Li, Eric C. Lai. "Diverse Cell-Specific Patterns of Alternative Polyadenylation in *Drosophila*" *Nature Communications* **13**(1), 5372 (2022)

Mehmet Neset Özel, Félix Simon, Shadi Jafari, Isabel Holguera, **Yen-Chung Chen**, Najate Benhra, Rana Naja El-Danaf, Katarina Kapuralin, Jennifer Amy Malin, Nikolaos Konstantinides, Claude Desplan. "Neuronal Diversity and Convergence in a Visual System Developmental Atlas" *Nature* **589**(7840), 88–95 (2021)

Yen-Chung Chen, Claude Desplan. "Gene Regulatory Networks during the Development of the *Drosophila* Visual System" *Current Topics in Developmental Biology* **139**, 89–125 (2020)

Ying-Tsen Tung, Kuan-Chih Peng, **Yen-Chung Chen**, Ya-Ping Yen, Mien Chang, Sebastian Thams, Jun-An Chen. "Mir-17~92 Confers Motor Neuron Subtype Differential Resistance to ALS-Associated Degeneration" *Cell Stem Cell* **25**(2), 193–209.e7 (2019)

Ya-Ping Yen, Wen-Fu Hsieh, Ya-Yin Tsai, Ya-Lin Lu, Ee Shan Liao, Ho-Chiang Hsu, **Yen-Chung Chen**, Ting-Chun Liu, Mien Chang, Joye Li, Shau-Ping Lin, Jui-Hung Hung, Jun-An Chen. "Dlk1-Dio3 Locus-Derived lncRNAs Perpetuate Postmitotic Motor Neuron Cell Fate and Subtype Identity" *eLife* **7**, e38080 (2018)

Presentation

66th Annual Drosophila Research Conference

San Diego, USA

SPEAKER AT WORKSHOP: SINGLE-CELL RESEARCH IN DROSOPHILA: TECHNOLOGIES AND RESOURCES

March 2025

- An efficient and universal single-cell transcriptomic analysis framework for cell-type-specific labeling and manipulation

Honors & Awards

2024	The Charlotte A. Pann Memorial Research Award , New York University	NY, USA
2023	Dean's Outstanding Graduate Student Teaching Award , New York University	NY, USA
2022	Training Program in Stem Cell and Regenerative Biology , NYSTEM	NY, USA
2022	Kopac Teaching Award II , New York University	NY, USA
2022	Chair's Graduate Fellowship , New York University	NY, USA
2022	Taiwanese Government Scholarship to Study Abroad , Ministry of Education	Taiwan
2017	Best Poster , International Conference of Developmental Biology, Stem Cells and Regenerative Medicine	Taiwan

Teaching and Work

R Bootcamp for Summer Undergraduate Research Program

New York University

INSTRUCTOR

Summer 2021, 2022, and 2023

A 10-session bootcamp for R in biology

Statistics in Biology

New York University

GRADUATE TEACHING ASSISTANT

Fall 2022

Recitation sessions of R and statistic methods

Sophomore Scholars Seminar

New York University

GRADUATE TEACHING ASSISTANT

Fall 2021 and Spring 2022

Discussion panels about evo-devo, immunology, gene editing, and art

Applied Genomics

New York University

GRADUATE TEACHING ASSISTANT

Spring 2021

Design and host recitation sessions for the use of modern genomic tools for alignment, variant calling, RNA-Seq, and ChIP-seq

Principles of Biology I/II

New York University

GRADUATE TEACHING ASSISTANT

Fall 2019 and Spring 2020

Manage recitation sections to help undergrads from a diverse field appreciate various aspects of life sciences